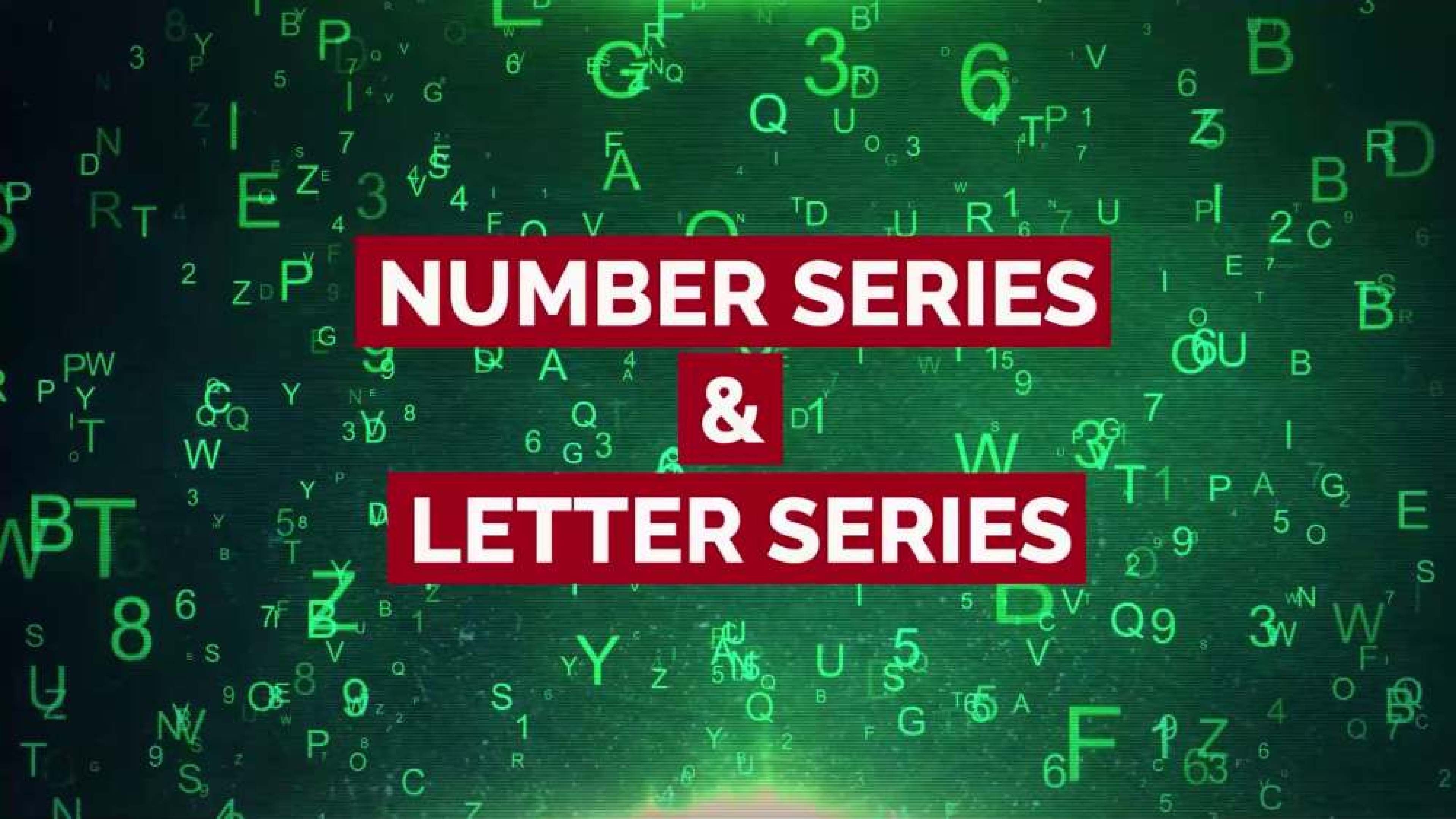


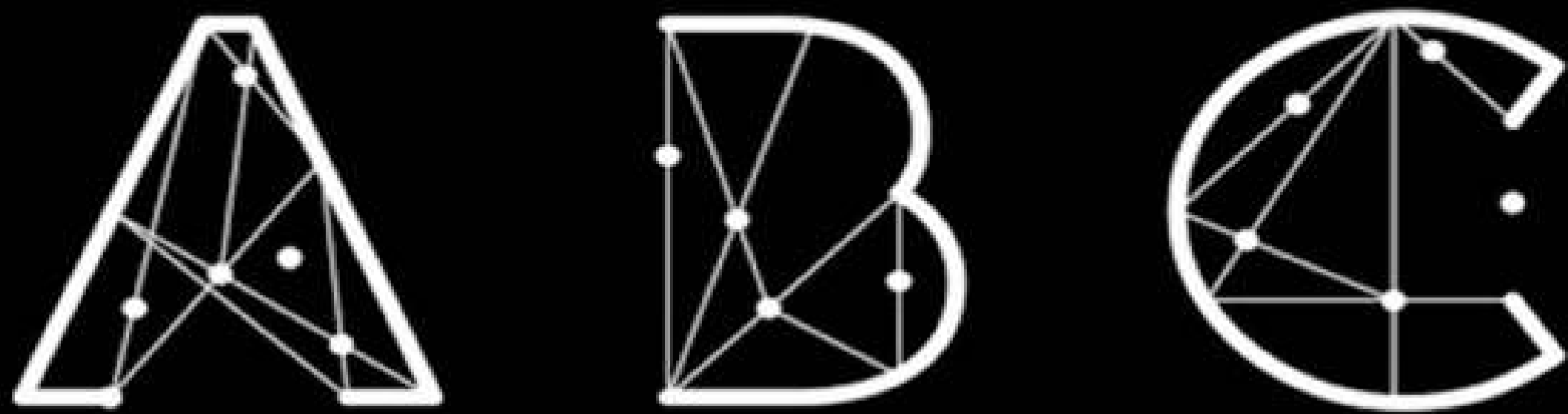
The background of the image is a dark green field filled with numerous small, light green characters. These characters include both uppercase and lowercase letters (A-Z) and digits (0-9), scattered randomly across the frame. Some characters are slightly larger or more prominent than others, creating a sense of depth and movement.

NUMBER SERIES

&

LETTER SERIES

NUMBER SERIES - LETTER SERIES



- by YASH JAIN

Solve This Number Puzzle

Find the Next Number in this series:

5, 7, 12, 19, 31, 50, ?



bAcK To basics



An Important Trick

2 5 10 17 26 37 —

2 6 12 20 30 42 —

0 2 6 12 20 30 —

An Important Trick

2 5 10 17 26 37 —

2 6 12 20 30 42 56
4 6 8 10 12 14

0 2 6 12 20 30 42
2 4 6 8 10 12

An Important Trick



2 6 12 20 30 42 —

0 2 6 12 20 30 —

An Important Trick

1^2	2^2	3^2	4^2	5^2	6^2	7^2
1	4	9	16	25	36	49 +1
2	5	10	17	26	37	<u>50</u> ✓
3	6	11	18	27	38	

✓

$\pm 1 \pm n$

2 6 12 20 30 42 —

0 2 6 12 20 30 —



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An Important Trick

2 5 10 17 26 37 —

$\pm n$

indexⁿ

0	1	2	3	4	5
1	2	3	4	5	6
2	6	12	20	30	42
					—

0 2 6 12 20 30 —



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An Important Trick

2 5 10 17 26 37 —

$$\begin{array}{c} \pm 14 \\ \pm n \end{array}$$

$\pm n$
index

3	8	15	24	35	48	7
2	6	12	20	30	42	56
± 1	± 2	± 3	± 4	± 5	± 6	
1	4	9	16	25	36	49

$-56 + 7 = 49$

0 2 6 12 20 30 —

An Important Trick

2 5 10 17 26 37 —

2 6 12 20 30 42 —

$\pm n$

1	2	3	4	5	6	7
0	2	6	12	20	30	42
1	4	9	16	25	36	49

Handwritten annotations: The numbers 7, 42, and 49 are circled in red. A red arrow points from the circled 42 to a circled $+1$ next to it. Another circled $+1$ is written above the circled 42.

Series with Combination

1. Square + 1
2. Square - 1
3. Cube + 1
4. Cube - 1
5. Square + n
6. Square - n
7. Cube + n
8. Cube - n
9. Cube + Square
10. Cube - Square



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Series with Combination

1. Square + 1

$$n^2 + 1$$

2. Square - 1

$$n^2 - 1$$

3. Cube + 1

$$n^3 + 1$$

4. Cube - 1

$$n^3 - 1$$

5. Square + n

$$n^2 + n$$

6. Square - n

$$n^2 - n$$

7. Cube + n

$$n^3 + n$$

8. Cube - n

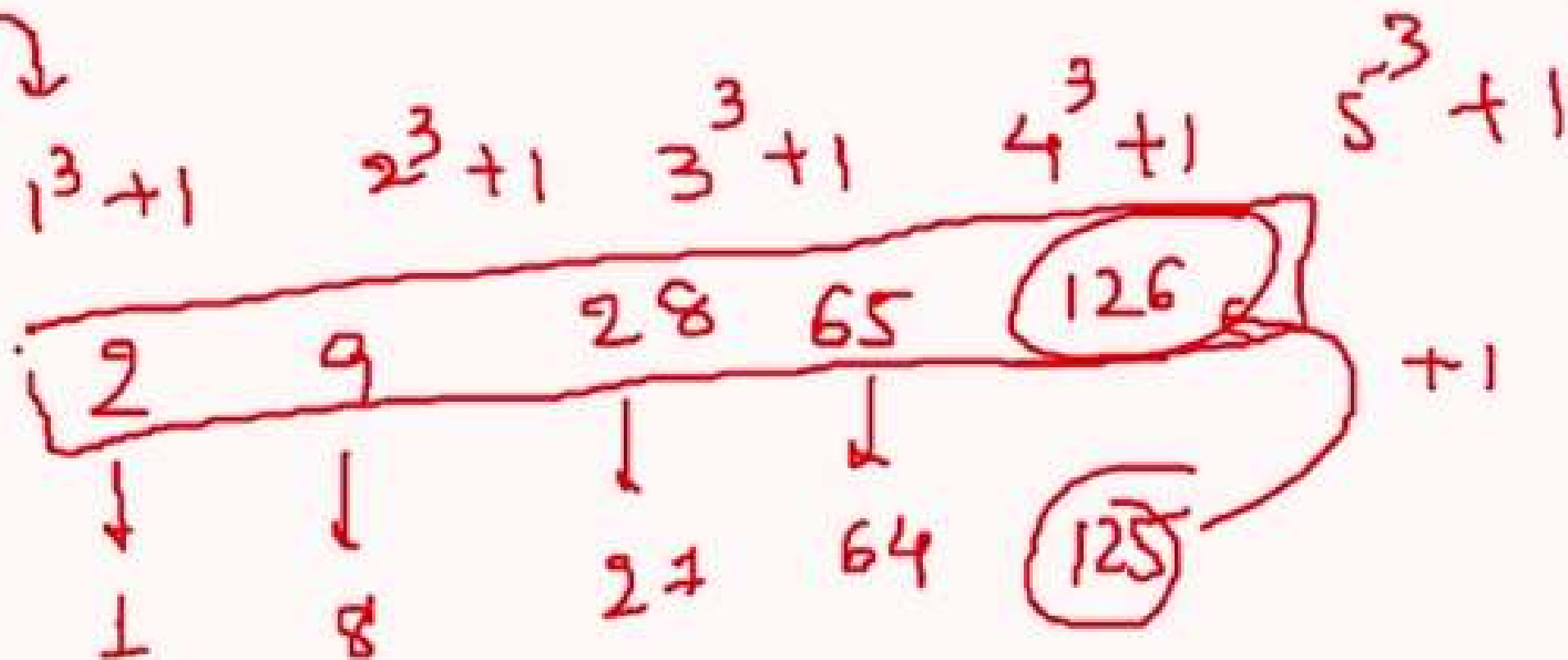
$$n^3 - n$$

9. Cube + Square

$$n^3 + n^2$$

10. Cube - Square

$$n^3 - n^2$$



An Important Trick

$2 \quad 5 \quad 10 \quad 17 \quad 26 \quad 37 \quad \text{---} \longrightarrow n^2 + 1$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $2^2+1 \quad 3^2+1 \quad 4^2+1 \quad 5^2+1 \quad 6^2+1$

$2 \quad 6 \quad 12 \quad 20 \quad 30 \quad 42 \quad \text{---}$

$\pm n$

$\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & (+1) \\ 0 & 2 & 6 & 12 & 20 & 30 & 42 \\ \textcircled{1} & \textcircled{4} & \textcircled{9} & \textcircled{16} & \textcircled{25} & \textcircled{36} & \textcircled{49} \end{array}$
 $\textcircled{42} \xrightarrow{+1} \textcircled{49}$

An Important Trick

$$\boxed{\pm n} \checkmark$$

2 5 10 17 26 37 ____ $\boxed{-1}$

$\boxed{2}$ $\boxed{6}$ $\boxed{12}$ $\boxed{20}$ $\boxed{30}$ 42 $\boxed{56}$ $n^2 + \boxed{n}$ $\boxed{\cancel{n}}$

2^2+2 4^2+4 6^2+6

1^2+1 3^2+3 5^2+5 7^2+7
 $+1$ $+2$ $+3$ $+4$ $+5$ $+6$ $+7$
 0 2 6 12 20 30 $\boxed{42}$

1^2-1 2^2-2 3^2-3 4^2-4 5^2-5 6^2-6

$7^2-7 = 49-7$
 n^2-n

Series with Combination

1. Square + 1

$$n^2 + 1$$

2. Square - 1

$$n^2 - 1$$

3. Cube + 1

$$n^3 + 1$$

4. Cube - 1

$$n^3 - 1$$

5. Square + n

$$n^2 + n$$

6. Square - n

$$n^2 - n$$

7. Cube + n

$$n^3 + n$$

8. Cube - n

$$n^3 - n$$

9. Cube + Square

$$n^3 + n^2$$

10. Cube - Square

$$n^3 - n^2$$

$1^3 + 1$ $2^3 + 1$ $3^3 + 1$ $4^3 + 1$ $5^3 + 1$ $1^3 - 1$ $2^3 - 1$ $3^3 - 1$ $4^3 - 1$ $5^3 - 1$

2	9	28	65	126
↓	↓	↓	↓	↓
1	8	27	64	125

$1^3 - 1^2$ $2^3 - 2^2$ $3^3 - 3^2$ $4^3 - 4^2$ $5^3 - 5^2$

Series with Combination

1. Square + 1

2. Square - 1

3. Cube + 1

4. Cube - 1

5. Square + n

6. Square - n

7. Cube + n

8. ~~Cube - n~~

9. Cube + Square ✓

10. Cube - Square ✓

$$\pm 1 \quad \pm n$$

Handwritten notes and diagrams:

- A box containing "Cube + n" and "~~Cube - n~~".
- Arrows point from these boxes to a circle containing "-n" and another circle containing "+n".
- A box contains the following expressions:
$$n^3 + n^2$$
$$n^3 - n^2$$